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Self-stopping tissue anchors

Abstract

A tissue anchor comprises an anchoring portion, and a crown coupled thereto. The crown includes an anchor head fixedly coupled to the anchoring portion, a driver interface, and a socket. The socket may be fixedly coupled to the driver interface and shaped to receive the anchor head. In a first state the anchor head is seated snugly within the socket, such that torque applied to the driver interface is transferred to the anchoring portion, thereby facilitating screwing of the anchoring portion into the tissue. Screwing of the anchoring portion into the tissue can pull the anchor head distally out of the socket, thereby transitioning the anchor into a second state in which torque applied to the driver interface rotates the socket relative to the anchor head and the anchoring portion. Other embodiments are also described.

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8628569	12/2013	Benichou et al.	N/A	N/A
8628571	12/2013	Hacohen et al.	N/A	N/A
8641727	12/2013	Starksen et al.	N/A	N/A
8652202	12/2013	Alon et al.	N/A	N/A
8652203	12/2013	Quadri et al.	N/A	N/A
8679174	12/2013	Ottma et al.	N/A	N/A
8685086	12/2013	Navia et al.	N/A	N/A
8728097	12/2013	Sugimoto et al.	N/A	N/A
8728155	12/2013	Montorfano et al.	N/A	N/A
8734467	12/2013	Miller et al.	N/A	N/A
8734699	12/2013	Heideman et al.	N/A	N/A
8740920	12/2013	Goldfarb et al.	N/A	N/A
8747463	12/2013	Fogarty et al.	N/A	N/A
8778021	12/2013	Cartledge	N/A	N/A
8784481	12/2013	Alkhatib et al.	N/A	N/A
8790367	12/2013	Nguyen et al.	N/A	N/A
8790394	12/2013	Miller et al.	N/A	N/A
8795298	12/2013	Hernlund et al.	N/A	N/A
8795355	12/2013	Alkhatib	N/A	N/A
8795356	12/2013	Quadri et al.	N/A	N/A
8795357	12/2013	Yohanan et al.	N/A	N/A
8808366	12/2013	Braido et al.	N/A	N/A
8808368	12/2013	Maisano et al.	N/A	N/A
8845717	12/2013	Khairkhahan et al.	N/A	N/A
8845723	12/2013	Spence et al.	N/A	N/A
8852261	12/2013	White	N/A	N/A
8852272	12/2013	Gross et al.	N/A	N/A
8858623	12/2013	Miller et al.	N/A	N/A
8864822	12/2013	Spence et al.	N/A	N/A
8870948	12/2013	Erzberger et al.	N/A	N/A
8870949	12/2013	Rowe	N/A	N/A
8888843	12/2013	Khairkhahan et al.	N/A	N/A
8889861	12/2013	Skead et al.	N/A	N/A
8894702	12/2013	Quadri et al.	N/A	N/A
8911461	12/2013	Traynor et al.	N/A	N/A
8911494	12/2013	Hammer et al.	N/A	N/A
8926696	12/2014	Cabiri et al.	N/A	N/A
8926697	12/2014	Gross et al.	N/A	N/A
8932343	12/2014	Alkhatib et al.	N/A	N/A
8932348	12/2014	Solem et al.	N/A	N/A
8940044	12/2014	Hammer et al.	N/A	N/A
8945211	12/2014	Sugimoto	N/A	N/A
8951285	12/2014	Sugimoto et al.	N/A	N/A
8951286	12/2014	Sugimoto et al.	N/A	N/A
8961595	12/2014	Alkhatib	N/A	N/A
8961602	12/2014	Kovach et al.	N/A	N/A
8979922	12/2014	Jayasinghe et al.	N/A	N/A
8992604	12/2014	Gross et al.	N/A	N/A
9005273	12/2014	Salahieh et al.	N/A	N/A

9011520	12/2014	Miller et al.	N/A	N/A
9011530	12/2014	Reich et al.	N/A	N/A
9023100	12/2014	Quadri et al.	N/A	N/A
9072603	12/2014	Tuval et al.	N/A	N/A
9107749	12/2014	Bobo et al.	N/A	N/A
9119719	12/2014	Zipory et al.	N/A	N/A
9125632	12/2014	Loulmet et al.	N/A	N/A
9125742	12/2014	Yoganathan et al.	N/A	N/A
9138316	12/2014	Bielefeld	N/A	N/A
9173646	12/2014	Fabro	N/A	N/A
9180005	12/2014	Lashinski et al.	N/A	N/A
9180007	12/2014	Reich et al.	N/A	N/A
9192472	12/2014	Gross et al.	N/A	N/A
9198756	12/2014	Aklog et al.	N/A	N/A
9226825	12/2015	Starksen et al.	N/A	N/A
9265608	12/2015	Miller et al.	N/A	N/A
9326857	12/2015	Cartledge et al.	N/A	N/A
9414921	12/2015	Miller et al.	N/A	N/A
9427316	12/2015	Schweich, Jr. et al.	N/A	N/A
9474606	12/2015	Zipory et al.	N/A	N/A
9526613	12/2015	Gross et al.	N/A	N/A
9561104	12/2016	Miller et al.	N/A	N/A
9579090	12/2016	Simms et al.	N/A	N/A
9693865	12/2016	Gilmore et al.	N/A	N/A
9730793	12/2016	Reich et al.	N/A	N/A
9788941	12/2016	Hacohen	N/A	N/A
9801720	12/2016	Gilmore et al.	N/A	N/A
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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS (1) The present application is a Continuation of International Patent Application PCT/IB2021/053314 to Brauon et al., filed Apr. 22, 2021, which published as WO 2021/255533, and which claims priority to Provisional U.S. Patent Application 63/041,423 to Brauon et al., filed Jun. 19, 2020, which is incorporated herein by reference.

BACKGROUND

(1) Tissue anchors are useable in a range of medical applications (e.g., for fastening an implant to tissue). Screwable tissue anchors can be configured to translate torque into distal motion, whereby the anchor (e.g., an anchoring portion thereof) is screwed into tissue.

SUMMARY OF THE INVENTION

(2) This summary is meant to provide some examples and is not intended to be limiting of the scope of the invention in any way. For example, any feature included in an example of this summary is not required by the claims, unless the claims explicitly recite the features. Also, the features, components, steps, concepts, etc. described in examples in this summary and elsewhere in this disclosure can be combined in a variety of ways. Various features and steps as described elsewhere in this disclosure can be included in the examples summarized here.

(3) Some applications herein are directed to tissue anchors that facilitate controlled anchoring of the tissue anchors into tissue of a subject, such as cardiac tissue. That is, the tissue anchors themselves include features that facilitate such control. For some applications, additional apparatus or system components(s), such as an anchor driver, is provided for screwing an anchoring portion of the tissue anchor into the tissue.

(4) For some applications, a tissue anchor limits a magnitude of torque which can be used to screw the anchor into tissue. For some applications, a tissue anchor limits a depth to which the tissue anchor can be screwed into tissue.

(5) Some tissue anchors described herein can comprise an anchoring portion, and a crown coupled thereto. The crown can include an anchor head, a driver interface, and a socket. The anchor head can be fixedly coupled to the anchoring portion. The socket can be fixedly coupled to the driver interface and can further be shaped to receive the anchor head. In a first state the anchor head is seated snugly within the socket, such that torque applied to the driver interface is transferred to the

anchoring portion, thereby facilitating screwing of the anchoring portion into the tissue. Screwing of the anchoring portion into the tissue pulls the anchor head distally out of the socket, thereby transitioning the anchor into a second state in which torque applied to the driver interface rotates the socket relative to the anchor head and the anchoring portion, e.g., such that the torque is no longer transferred to the anchoring portion, e.g., such that further screwing of the anchoring portion into the tissue is not possible.

(6) Some tissue anchors described herein can comprise a crown including an anchor head that is fixedly coupled to the anchoring portion. The crown can define a driver interface configured to be engaged by a driver. The crown can also define a tissue-facing surface, such that screwing the anchoring portion into the tissue moves the tissue-facing surface distally toward the tissue.

(7) For some applications, the crown can comprise a socket that can be fixedly coupled to the driver interface and is shaped to receive the anchor head. For some such applications, the tissue anchor can have (i) a torque-transfer state in which the anchor head is seated within the socket, such that torque applied by the driver to the driver interface rotates the socket, the anchor head, and the anchoring portion, thereby screwing the anchoring portion into the tissue, and (ii) a non-torque-transfer state in which the anchor head is disposed distally from the socket, such that applied torque is not transferred from the interface to the anchor head and the anchoring portion (or is at least significantly reduced).

(8) For such applications, contact between the tissue-facing surface and the tissue can increase resistance against further distal movement of the tissue-facing surface. At this stage, further screwing of the anchor can pull the anchor head distally out of the socket (e.g., towards the tissue-facing surface), such that the tissue anchor transitions from the torque-transfer state to the non-torque-transfer state, thereby limiting the distal force applied by the tissue-facing surface to the tissue.

(9) For some such applications, the tissue anchor can further comprise a spring disposed between the anchor head and the tissue-facing surface. Pulling the anchor head distally out of the socket and towards the tissue-facing surface can compress the spring between the anchor head and the tissue-facing surface, facilitating continued screwing of the anchoring portion into the tissue while the tissue anchor transitions from the torque-transfer state to the non-torque-transfer state.

(10) For some applications, the crown can comprise a slip clutch that transfers torque from the driver interface to the anchor head, while limiting the transferred torque to not exceed a torque threshold.

(11) For some applications in which the crown comprises a slip clutch, the slip clutch can transfer torque from the driver interface to the anchor head via a cantilever pin that revolves, with the anchor head, around the longitudinal axis of the anchor, while a torque-applying portion of the pin remains in contact with a noncircular lateral surface of the anchor head. However, when the applied torque exceeds the torque threshold, the torque-applying portion of the pin deflects away from the anchor head (e.g., due to being pushed laterally outward by the geometry of the anchor head), such that the driver interface and the pin can rotate with respect to the anchor head and the anchoring portion, thereby limiting the transferred torque.

(12) For some such applications, deflection of the torque-applying portion of the pin away from the anchor head is dependent upon whether the applied torque is forward torque or reverse torque. For example, application of forward torque exceeding the torque threshold can cause the pin to deflect. However, application of reverse torque exceeding the torque threshold may not cause the pin to deflect. For example, the pin may revolve in a reverse rotational direction while the pin remains in contact with the anchor head, causing the anchoring portion to unscrew from the tissue. It is hypothesized that, in this way, the anchor can limit torque for screwing of the anchor into the tissue, while reliably allowing sufficient torque for unscrewing of the anchor from the tissue.

(13) For some applications in which the crown comprises a slip clutch, the anchor head comprises a gear shaped to define a lateral surface and a notch. For such applications, the slip clutch transfers

torque to the gear via a cantilever pin that revolves, with the anchor head, around the longitudinal axis of the anchor, while a torque-applying portion of the pin remains in contact with the gear. However, when the applied torque exceeds the torque threshold, the torque-applying portion of the pin deflects away from the gear (e.g., due to being pushed laterally outward by the geometry of the anchor head), such that the driver interface and the pin rotate with respect to the gear and the anchoring portion, thereby limiting the transferred torque.

(14) For some such applications, deflection of the torque-applying portion of the pin away from the gear is dependent upon whether the torque is applied to the interface is forward torque or reverse torque. For example, application of forward torque exceeding the torque threshold can cause the pin to deflect away from the gear. However, application of reverse torque exceeding the torque threshold may not cause the pin to deflect. Instead, the pin can revolve in a reverse direction until an end-portion of the pin is latched into the notch defined by the gear. In this way, application of reverse torque causes the gear and the anchoring portion to rotate with the driver interface, causing the anchoring portion to unscrew from the tissue. It is hypothesized that, in this way, the anchor can limit torque for screwing of the anchor into the tissue, while reliably allowing sufficient torque for unscrewing of the anchor from the tissue.

(15) There is therefore provided, in accordance with some applications, a system and/or apparatus for use with tissue of a subject, the system/apparatus including a driver and a tissue anchor, the tissue anchor including an anchoring portion configured to be screwed distally into the tissue by being rotated about a longitudinal axis of the anchor and a crown, coupled to a proximal portion of the anchoring portion, defining a tissue-facing surface.

(16) In some applications, the tissue anchor and/or crown of the tissue anchor includes an anchor head fixedly coupled to the anchoring portion, such that screwing the anchoring portion into the tissue moves the anchor head distally along the longitudinal axis toward the tissue and/or a driver interface, configured to be engaged by the driver.

(17) In some applications, the system/apparatus includes a socket, fixedly coupled to the driver interface, and shaped to receive the anchor head within the socket, the tissue-facing surface facing distally away from the socket.

(18) In some applications, the system/apparatus or tissue anchor thereof has a first state in which the anchor head is seated snugly within the socket, such that torque applied by the driver to the driver interface rotates the socket, the anchor head, and the anchoring portion, thereby facilitating screwing of the anchoring portion into the tissue.

(19) In some applications, the system/apparatus or tissue anchor thereof has a second state in which the anchor head is disposed distally from the socket, such that torque applied by the driver to the driver interface rotates the socket relative to the anchor head and the anchoring portion.

(20) In some applications, the tissue anchor is configured to transition from the first state to the second state, responsively to the anchoring portion having been screwed into the tissue sufficiently deep such that the tissue resists further distal movement of the tissue-facing surface while the screwing of the anchoring portion into the tissue pulls the anchor head distally out of the socket.

(21) In some applications, the driver interface defines a floor that separates the driver from the anchor head while the driver interface is engaged by the driver.

(22) In some applications, the anchor head is shaped such that a transverse cross-section of the anchor head defines a non-circular profile. In some applications, the anchor head is shaped such that the transverse cross-section of the anchor head defines a plurality of lateral surfaces. In some applications, the anchor head is shaped such that the transverse cross-section of the anchor head defines a polygon. In some applications, the anchor head is shaped such that the transverse cross-section of the anchor head defines a square. In some applications, the anchor head is shaped such that the transverse cross-section of the anchor head defines a hexagon.

(23) In some applications, the tissue is tissue of a heart of the subject, and the tissue anchor is transluminally advanceable to the heart.

- (24) In some applications, the driver includes a flexible shaft and a driver head at a distal end of the shaft, such that the anchor driver is transluminally advanceable to the heart.
- (25) In some applications, the crown includes a casing, the casing dimensioned to define: the driver interface, the socket, the tissue-facing surface, and a free zone disposed between the socket and the tissue-facing surface, and while the anchor is in the second state, the anchor head is disposed within the free zone.
- (26) In some applications, the anchor head is configured to rotate with respect to the socket while the anchor head is disposed in the free zone.
- (27) In some applications, the driver includes a driver head, the driver head shaped to define a shoulder, the shoulder: positioned on a side of the driver head, and dimensioned such that, while the driver interface is engaged by the driver head, the shoulder contacts a proximal surface of the casing.
- (28) In some applications, the system/apparatus includes a spring disposed within the casing, between the anchor head and the tissue-facing surface, and the anchor is configured such that while the anchor transitions from the first state to the second state: screwing the anchoring portion into the tissue pulls the anchor head distally out of the socket, compressing the spring.
- (29) In some applications, the anchor is configured such that while the anchor transitions from the first state to the second state, screwing the anchoring portion into the tissue pulls the anchor head distally out of the socket, compressing the spring and pressing the tissue-facing surface against the tissue.
- (30) In some applications, the anchor is configured such that while the anchor transitions from the first state to the second state, screwing the anchoring portion into the tissue pulls the anchor head distally out of the socket, compressing the spring while the anchor head is: partially disposed within the socket, and partially disposed within the free zone.
- (31) The system and/or apparatus can further comprise an implant, and the tissue anchor can be configured to secure the implant to the tissue. In some applications, the implant comprises a tether or contraction member. In some applications, the tissue anchor is configured to secure the tether or contraction member to the tissue. In some applications, the tissue anchor is configured to secure the tether or contraction member to the tissue such that applying tension to the tether or contraction member changes a shape and/or size of the tissue.
- (32) There is further provided, in accordance with some applications, a system and/or apparatus including: a driver, including a shaft and a driver head at a distal end of the shaft; and a tissue anchor. The tissue anchor includes an anchoring portion configured to be screwed distally into the tissue by being rotated about a longitudinal axis of the anchor.
- (33) In some applications, the tissue anchor includes a crown, coupled to a proximal portion of the anchoring portion.
- (34) In some applications, the tissue anchor and/or crown includes an anchor head fixedly coupled to the anchoring portion, such that rotation of the anchor head rotates the anchoring portion about the longitudinal axis.
- (35) In some applications, the tissue anchor, crown, and/or anchor head includes a driver interface, configured to be engaged by the driver head and rotated by the driver.
- (36) In some applications, the tissue anchor, crown, and/or anchor head includes a slip clutch. In some implementations, the slip clutch is coupled to the driver interface and/or to the anchor head. In some implementations, the slip clutch is configured to (i) transfer, to the anchor head, torque applied to the driver interface, up to a torque threshold, and to (ii) slip in response to torque greater than the torque threshold applied to the driver interface, thereby limiting torque transferred to the anchor head to not exceed the torque threshold.
- (37) In some applications, the anchor head is shaped to define a non-circular lateral surface, the slip clutch includes a cantilever pin, a portion of the cantilever pin fixedly coupled to the driver interface. In some implementations, the slip clutch is configured: (i) to transfer torque from the

driver interface to the anchor head by revolving about the longitudinal axis in response to the driver interface rotating while a torque-applying portion of the pin is in contact with the non-circular lateral surface of the anchor head, and (ii) to slip by the pin being deflected away from the longitudinal axis by the anchor head.

(38) In some applications, the slip clutch includes a cantilever pin, a portion of the cantilever pin fixedly coupled to the driver interface, and the slip clutch is configured to slip in response to torque greater than the torque threshold being applied to the driver interface, by the pin being deflected away from the longitudinal axis by the anchor head.

(39) In some applications, the tissue is tissue of a heart of a subject, and the tissue anchor is transluminally advanceable to the heart.

(40) In some applications, the driver includes a flexible shaft and a driver head at a distal end of the shaft, such that the anchor driver is transluminally advanceable to the heart.

(41) In some applications, the slip clutch is configured to selectively rotationally couple the driver interface to the anchor head, such that: (i) in response to application, to the driver interface, of torque in a first rotational direction and at a first magnitude that does not exceed the torque threshold, the anchor head and the anchoring portion rotate with the driver interface in the first rotational direction; in response to application, to the driver interface, of torque in the first rotational direction and at a second magnitude that exceeds the torque threshold, the slip clutch slips such that the driver interface rotates with respect to the anchor head and the anchoring portion; and in response to application, to the driver interface, of torque in a second rotational direction and at the second magnitude, the anchor head and the anchoring portion rotate with the driver interface in the second rotational direction, the second rotational direction being opposite to the first rotational direction.

(42) In some applications, the torque threshold is a first torque threshold, and the tissue anchor is configured such that application, to the driver interface, of torque in the second rotational direction and at a third magnitude exceeding a second torque threshold that is greater than the first torque threshold causes the slip clutch to slip such that the driver interface rotates with respect to the anchor head and the anchoring portion.

(43) In some applications, the anchoring portion is oriented with respect to the slip clutch such that: rotation of the anchor head and the anchoring portion in the first rotational direction facilitates screwing of the tissue anchor into the tissue, and rotation of the anchor head and the anchoring portion in the second rotational direction facilitates unscrewing of the tissue anchor from the tissue.

(44) In some applications, the anchor head is shaped to define a non-circular lateral surface, the slip clutch includes a cantilever pin, a fixed portion of the cantilever pin fixedly coupled to the driver interface. In some implementations, the slip clutch is configured: to transfer torque from the driver interface to the anchor head by revolving about the longitudinal axis in response to the driver interface rotating while a torque-applying portion of the pin is in contact with the non-circular lateral surface of the anchor head, and to slip by the pin being deflected away from the longitudinal axis by the anchor head.

(45) In some applications, the slip clutch is configured such that application of torque to the driver interface, in the first rotational direction and at the first magnitude, causes the anchor head and the anchoring portion to rotate with the driver interface, by the cantilever pin revolving about the longitudinal axis in the first rotational direction with the torque-applying portion ahead of the fixed portion.

(46) In some applications, the slip clutch is configured such that application of torque to the driver interface, in the first rotational direction and at the first magnitude, causes the anchor head and the anchoring portion to rotate with the driver interface, by the cantilever pin revolving about the longitudinal axis in the first rotational direction with the torque-applying portion trailing the fixed portion.

(47) In some applications, the slip clutch is configured such that: the torque-applying portion is a

first torque-applying portion, and application of torque to the driver interface, in the first rotational direction and at the first magnitude, causes the anchor head and the anchoring portion to rotate with the driver interface, by the cantilever pin revolving about the longitudinal axis in the first rotational direction while the first torque-applying portion is in contact with the non-circular lateral surface of the anchor head. In some implementations, application of torque, to the driver interface, in the first rotational direction and at the second magnitude, causes the driver interface and the pin to rotate in the first rotational direction, with respect to the anchor head and the anchoring portion, by the pin being deflected away from the longitudinal axis by the anchor head. In some implementations, application of torque, to the driver interface, in the second rotational direction and at the second magnitude, causes the anchor head and the anchoring portion to rotate with the driver interface in the second rotational direction, by the cantilever pin revolving about the longitudinal axis in the second rotational direction while a second torque-applying portion of the pin is in contact with the anchor head.

(48) In some applications, the slip clutch is configured such that: while torque is applied to the driver interface in the first rotational direction and at the first magnitude, the pin has a forward cantilever span between (i) the fixed portion of the pin, and (ii) the first torque-applying portion, and while torque is applied to the driver interface in the second rotational direction, the pin has a reverse cantilever span between (i) the fixed portion of the pin, and (ii) the second torque-applying portion, the forward cantilever span being longer than the reverse cantilever span.

(49) In some applications, the first torque-applying portion is further than the second torque-applying portion from the fixed portion. In some applications, the second torque-applying portion is further than the first torque-applying portion from the fixed portion.

(50) In some applications: the anchor head is shaped to define a notch, the pin defines a pawl, which serves as the second torque-applying portion, and the slip clutch is configured such that application of torque to the driver interface in the second rotational direction and at the second magnitude, causes the anchor head and the anchoring portion to rotate with the driver interface in the second rotational direction, by the cantilever pin revolving about the longitudinal axis in the second rotational direction while the pawl is latched into the notch defined by the anchor head.

(51) In some applications, the slip clutch is configured such that application of torque to the driver interface in the second rotational direction and at the second magnitude, causes the driver interface and the pin to rotate in the second rotational direction, with respect to the anchor head and the anchoring portion, for not more than a quarter turn, before the pawl becomes latched into the notch defined by the anchor head.

(52) The system and/or apparatus can further comprise an implant, and the tissue anchor can be configured to secure the implant to the tissue. In some applications, the implant comprises a tether or contraction member. In some applications, the tissue anchor is configured to secure the tether or contraction member to the tissue. In some applications, the tissue anchor is configured to secure the tether or contraction member to the tissue such that applying tension to the tether or contraction member changes a shape and/or size of the tissue.

(53) There is further provided, in accordance with some applications, a method for use with a tissue of a subject includes advancing a tissue anchor to the tissue, the tissue anchor including an anchoring portion, anchor head, and a driver interface, and screwing the anchoring portion distally into the tissue by applying torque to the driver interface, such that the driver interface and the anchoring portion rotate together about a longitudinal axis of the anchor, continuing to screw the anchoring portion distally into the tissue, at least until the driver interface becomes rotatable relative to the anchor head.

(54) In some applications, the tissue anchor includes a crown coupled to a proximal portion of the anchoring portion, and the crown can include the anchor head, which can be fixedly coupled to the anchoring portion. The driver interface can also be part of the crown in some implementations.

(55) The method can further include engaging a driver (e.g., an anchor driver, etc.) with the driver

interface. The method can further include using the driver to screw the anchoring portion distally into the tissue by applying torque to the driver interface, such that the driver interface, the anchor head, and the anchoring portion rotate together about a longitudinal axis of the anchor.

(56) The method can further include, subsequently to screwing the anchor, disengaging the driver from the driver interface and removing the driver from the subject, while leaving the tissue anchor anchored to the tissue.

(57) In some applications, the driver includes a driver head, the driver head shaped to define a shoulder, and screwing the anchoring portion distally into the tissue by applying torque to the driver interface includes screwing the anchoring portion distally into the tissue by applying torque to the driver interface while: the driver head is engaged with the driver interface, and the shoulder is in contact with a proximal surface of the crown.

(58) In some applications, the driver interface defines a floor, and screwing the anchoring portion distally into the tissue by applying torque to the driver interface includes, using the driver head, contacting the floor.

(59) In some applications, the tissue is tissue of a heart of the subject and advancing the tissue anchor to the tissue includes transluminally advancing the tissue anchor to the heart. In some implementations, the tissue is tissue of a mitral valve or a tricuspid valve of a heart.

(60) In some applications, the driver includes a flexible shaft and a driver head at a distal end of the shaft, and the method includes, subsequently to advancing the tissue anchor to the tissue, using the flexible shaft, transluminally advancing the driver head to the driver interface.

(61) In some applications, the crown is shaped to define: a socket fixedly coupled to the driver interface, and a tissue-facing surface facing distally away from the socket; and screwing the anchoring portion distally into the tissue includes screwing the anchoring portion distally into the tissue by applying torque to the driver interface while the anchor head is seated snugly within the socket, such that the driver interface, the socket, the anchor head, and the anchoring portion rotate about the longitudinal axis; and continuing to screw the anchoring portion distally into the tissue includes continuing to screw the anchoring portion distally into the tissue at least until: the tissue resists further distal movement of the tissue-facing surface, and the anchor head becomes pulled distally out of the socket, such that the socket becomes rotatable relative to the anchor head.

(62) In some applications, the crown and/or anchor head is shaped to define a free zone between the socket and the tissue-facing surface, and continuing to screw the anchoring portion distally into the tissue includes continuing to screw the anchoring portion distally into the tissue at least until the anchor head becomes pulled distally into the free zone.

(63) In some applications, continuing to screw the anchoring portion distally into the tissue includes continuing to screw the anchoring portion distally into the tissue at least until the anchor head becomes pulled entirely into the free zone.

(64) In some applications, the tissue anchor includes a spring, the spring disposed between the anchor head and the tissue-facing surface, and continuing to screw the anchoring portion distally into the tissue includes continuing to screw the anchoring portion distally into the tissue at least until the anchor head compresses the spring.

(65) In some applications, continuing to screw the anchoring portion distally into the tissue includes continuing to screw the anchoring portion distally into the tissue at least until the spring presses the tissue-facing surface against the tissue.

(66) In some applications, the crown includes a slip clutch, the slip clutch being coupled to the driver interface and to the anchor head, screwing the anchoring portion distally into the tissue includes screwing the anchoring portion distally into the tissue by applying torque to the driver interface, such that torque is transferred from the driver interface, via the slip clutch, to the anchor head, and continuing to screw the anchoring portion distally into the tissue includes continuing to screw the anchoring portion distally into the tissue at least until, at a torque threshold, the slip clutch slips, such that the driver interface rotates with respect to the anchor head and the anchoring

portion.

(67) In some applications, the anchor head is shaped to define a non-circular lateral surface, the slip clutch includes a cantilever pin, a fixed portion of the cantilever pin being fixedly coupled to the driver interface, screwing the anchoring portion distally into the tissue includes screwing the anchoring portion distally into the tissue by applying torque to the driver interface, such that the driver interface and the pin rotate about the longitudinal axis in a first rotational direction, together with the anchor head and the anchoring portion, while a torque-applying portion of the pin presses against the lateral surface of the anchor head. In some implementations, continuing to screw the anchoring portion distally into the tissue includes continuing to screw the anchoring portion distally into the tissue at least until, at the torque threshold, the slip clutch slips by the anchor head deflecting the pin away from the longitudinal axis, such that the driver interface rotates with respect to the anchor head and the anchoring portion.

(68) In some applications, the torque-applying portion of the pin is a first torque-applying portion of the pin, screwing the anchoring portion distally into the tissue includes screwing the anchoring portion distally into the tissue by applying torque to the driver interface, such that the driver interface and the pin rotate about the longitudinal axis in the first rotational direction, together with the anchor head and the anchoring portion, while the first torque-applying portion of the pin presses against the lateral surface of the anchor head. In some implementations, continuing to screw the anchoring portion distally into the tissue includes continuing to screw the anchoring portion distally into the tissue at least until, at the torque threshold, the slip clutch slips by the anchor head pushing against the first torque-applying portion and thereby deflecting the pin away from the longitudinal axis, such that the driver interface rotates with respect to the anchor head and the anchoring portion.

(69) In some applications, the method includes unscrewing the anchoring portion proximally from the tissue, by applying torque in a second rotational direction to the driver interface, such that the anchor head and the anchoring portion rotate with the driver interface, in the second rotational direction while a second torque-applying portion of the pin presses against the anchor head.

(70) In some applications, the anchor head is shaped to define a notch, the pin defines a pawl that serves as the second torque-applying portion, and unscrewing the anchoring portion proximally from the tissue includes unscrewing the anchoring portion proximally from the tissue, by applying torque, in the second rotational direction, to the driver interface, such that the anchor head and the anchoring portion rotate with the driver interface, in the second rotational direction while the pawl is latched into the notch.

(71) In some applications, the methods herein further comprise anchoring an implant, tether and/or contraction member to the tissue. The tissue anchor can be configured to anchor or secure the implant, tether, and/or contraction member to the tissue. In some applications, the tissue anchor is used to anchor or secure the implant, tether, and/or contraction member to the tissue.

(72) Anchoring or securing the implant, tether, and/or contraction member to the tissue can comprise screwing the anchoring portion distally into the tissue by applying torque to the driver interface, such that the driver interface and the anchoring portion rotate together about a longitudinal axis of the anchor, and continuing to screw the anchoring portion distally into the tissue, at least until the driver interface becomes rotatable relative to the anchor head.

(73) In some applications, after anchoring or securing the implant, tether, and/or contraction member to the tissue, the method further comprises applying tension to the tether or contraction member to change a shape and/or size of the tissue (e.g., to change a shape and/or size of an annulus of a heart valve, etc.)

(74) The above method(s) can be performed on a living animal or on a simulation, such as on a cadaver, cadaver heart, simulator (e.g., with the body parts, heart, tissue, etc. being simulated), etc.

(75) The present invention will be more fully understood from the following detailed description of applications thereof, taken together with the drawings, in which:

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. **1A-E**, **2A-H**, and **3A-H** are schematic illustrations showing example tissue anchor systems and their use, in accordance with some applications;
- (2) FIGS. **4A-D**, **5A-E** and **6A-B** are schematic illustrations showing an example tissue anchor system, in accordance with some applications; and
- (3) FIGS. **7A-C**, **8A-E** and **9A-C** are schematic illustrations showing an example tissue anchor system, in accordance with some applications.

DETAILED DESCRIPTION

- (4) Reference is made to FIGS. **1A-E**, **2A-H**, and **3A-H**, which are schematic illustrations showing use of an example tissue anchor system **10**, in accordance with some applications.
- (5) System **10** comprises a tissue anchor **12**, and an anchor driver **40**. As shown, anchor **12** comprises an anchoring portion (i.e., a tissue-engaging element) **34** shaped to facilitate screwing of the anchoring portion into tissue **90** by being rotated about a longitudinal axis ax1 of anchor **12**. For example, and as shown, anchoring portion **34** is shaped as a corkscrew having a distal tissue-piercing point **36**. This is not meant to exclude other shapes which facilitate anchoring portion **34** being screwed into tissue **90**. For example, anchoring portion can be shaped to define a threaded shank.
- (6) In some applications, and as shown, anchoring portion **34** is coupled, at a proximal portion of the anchoring portion, to a crown **38** that defines a tissue-facing surface **28**. FIGS. **1A-C** are side-views that show crown **38** in longitudinal cross-section (but anchoring portion **34** is shown whole). FIG. **1D** is a top-view of anchor **12**, i.e., looking down at crown **38**. FIG. **1E** is a transverse cross-section through crown **38**, at the level indicated in FIG. **1B**. As shown, crown **38** comprises an anchor head **30** and a socket **24** shaped to receive the anchor head, e.g., by the anchor head being reversibly seated within the socket. FIG. **1B** shows anchor head **30** seated within socket **24**, and FIG. **1C** shows the anchor head having exited the socket. For some such applications, and as shown in cross-section in FIG. **1E**, socket **24** and anchor head **30** are shaped complementarily to each other, such that the socket snugly receives the anchor head.
- (7) In some applications, and as shown, crown **38** comprises a driver interface **41** coupled to socket **24** such that the interface and the socket are rotationally fixed. Driver **40** is configured to engage the driver interface **41**. For some applications, driver **40** is transluminally advanced to tissue **90** of a subject (e.g., tissue of a heart of the subject), e.g., while coupled to anchor **12**. For some such applications, driver **40** having a shaft **44** (e.g., a flexible shaft) facilitates transluminal advancement of the driver to tissue **90**.
- (8) In system **10**, torque is transferred indirectly from driver **40** to anchor head **30**, e.g., via driver interface **41** and socket **24**. For example, driver **40** can comprise a driver head **45** at a distal end of shaft **44**, that is reversibly seatable within driver interface **41** (FIG. **1A**). For some applications, and as shown, driver **40** (e.g., driver head **45**) does not contact anchor head **30** while driver head **45** is seated within driver interface **41**. For example, anchor head **30** can be inaccessible to driver head **45** due to the anchor head being enclosed in a casing **20**, e.g., as described hereinbelow.
- (9) In some applications, tissue-facing surface **28** faces distally away from socket **24**. In some applications, tissue-facing surface **28** is axially fixed in relation to driver interface **41**, such that screwing tissue anchor **12** into tissue **90** typically brings surface **28** closer to the tissue (e.g., brings the tissue-facing surface into contact with the tissue). For some applications, tissue-facing surface **28** is also rotationally fixed with respect to driver interface **41** (e.g., tissue-facing surface is fixedly attached to driver interface **41**).
- (10) For some applications, and as shown, anchor **12** (e.g., crown **38** thereof) comprises casing **20** that comprises socket **24**, interface **41** and surface **28**. Casing **20** can be a unitary structure that is

shaped to define socket **24**, interface **41** and surface **28**. Casing **20** can house anchor head **30**, such that crown **38** comprises both the anchor head and the casing that houses the anchor head. Often for such applications, casing **20** is dimensioned to define a free zone **25** within which anchor head **30** is disposed while the anchor head is not seated in socket **24** (FIG. **1C**). Free zone **25** can be disposed distally from socket **24** (i.e., closer than socket **24** to anchoring portion **34**). Although socket **24** is configured to transfer torque from interface **41** to anchor head **30** while the anchor head is disposed in the socket (e.g., due to the snug fit therebetween), free zone **25** is configured to allow the socket to rotate with respect to the anchor head while the anchor head is disposed in the free zone and not in the socket (e.g., such that torque is not transferred from the socket to the anchor head).

(11) For some applications, socket **24**, interface **41**, and surface **28** of casing **20** are rotationally fixed in relation to each other, such that rotation of one part of the casing rotates the entire casing. This is not meant to exclude applications in which tissue-facing surface **28**, socket **24** and/or interface **41** are discrete elements.

(12) As described hereinabove, system **10** can be configured such that anchor head **30** is inaccessible to driver **40** (e.g., to anchor head **45** thereof). For some applications, interface **41** (or another component of crown **38**) defines a floor **42**, which separates driver **40** from anchor head **30** while the driver is seated within the interface. Alternatively or additionally, driver **40** (e.g., driver head **45** thereof) is shaped to define one or more shoulders **46** (e.g., positioned laterally, as shown in FIG. **1A**). Often for such applications, shoulders **46** are dimensioned such that, while driver interface **41** is engaged by driver **40** (e.g., while driver head **45** is seated within the interface), the shoulders contact a proximal surface **29** of casing **20**. Anchor head **30** being inaccessible to driver **40** facilitates transfer of torque from driver **40** to driver interface **41**, while reducing (e.g., eliminating) direct transfer of a pushing force from the driver to anchor head **30**, e.g., restricting any transfer of the pushing force to be via casing **20**.

(13) In some applications, and as shown, anchor **12** is transitionable between a first state (e.g., a torque-transfer state, FIG. **1B**) and a second state (e.g., a non-torque-transfer state, FIG. **1C**). In some applications, and as shown, while anchor **12** is in the first state, anchor head **30** is seated within socket **24**. In this way, torque applied by driver **40** to driver interface **41** rotates socket **24**, anchor head **30**, and anchoring portion **34**. In some applications, and as described hereinbelow, the first state facilitates screwing anchoring portion **34** into tissue **90**.

(14) Further in some applications and as shown, while anchor **12** is in the second state, anchor head **30** is disposed outside of (e.g., distally from) socket **24**, such that torque applied by driver **40** to driver interface **41** rotates socket **24** relative to anchor head **30** (and thereby anchoring portion **34**), e.g., such that torque is not transferred from the driver to the anchor head and the anchoring portion. Transition of anchor **12** from the first state to the second state occurs in response to anchor head **30** being pulled distally out of socket **24** by screwing of the anchor into tissue **90**, e.g., as described hereinbelow in reference to FIGS. **2A-H**.

(15) For some applications, and as shown, anchor **12** further comprises a compression spring **26** that can be disposed between anchor head **30** and tissue-facing surface **28** (e.g., within casing **20**). The function of spring **26** is described in more detail hereinbelow.

(16) FIGS. **2A-H** show driver **40** being used to screw anchoring portion **34** of anchor **12** into tissue **90** while anchor head **30** moves distally along longitudinal axis **ax1**, pressing surface **28** against the tissue. FIGS. **3A-H** show corresponding steps, but with a variant **12'** of anchor **12** that does not comprise compression spring **26**. Henceforth, this variant of anchor **12** is referred to as anchor **12'**.

(17) FIG. **2A** shows anchor **12** disposed against tissue **90**, such that distal tissue-piercing point **36** contacts the tissue. Torque is then applied from driver head **45**, via interface **41** and socket **24**, to anchor **12**, screwing anchoring portion **34** into tissue **90** (FIG. **2B**). During this time, anchor **12** is in its torque-transfer state, and typically behaves similarly to a prior art tissue anchor of unitary construction. Thus, screwing of anchoring portion **34** into tissue **90** results in tissue-facing surface

28 moving distally along the longitudinal axis, until it contacts the tissue (FIG. 2C).

(18) FIGS. 2D-F show further screwing of anchoring portion **34** into tissue **90**, by continued rotation of driver head **45** and crown **38**, despite anchoring portion **34** having already been screwed into tissue **90** sufficiently deep such that the tissue resists further distal movement of tissue-facing surface **28** and socket **24** (and often the entirety of casing **20**). As shown in FIGS. 2D-F, this resistance contributes to the transition of the anchor from the first state to the second state, because once surface **28** has made contact with tissue **90** further screwing of anchoring portion **34** into the tissue progressively pulls anchor head **30** distally out of socket **24**, which is inhibited by tissue **90** from advancing further distally. Since resistance from tissue **90** contributes to the transition of anchor **12** to the second state, torque applied to driver interface **41** is translated into distal motion of anchor head **30** relative to tissue-facing surface **28** (e.g., within casing **20**), instead of being translated into application of a distal force by tissue-facing surface **28** to tissue **90**. It is therefore hypothesized that triggering transition of anchor **12** from the first state to the second state, by tissue **90** resisting further distal movement of tissue-facing surface **28**, may advantageously limit: (i) the distal force applied by the tissue-facing surface to the tissue while the anchor is screwed into the tissue, and/or (ii) a depth to which the tissue anchor can be screwed into the tissue.

(19) In FIGS. 2D-E, anchor head **30** has moved further distally within casing **20**, but has not yet completely exited socket **24** (e.g., the anchor head is partially disposed within the socket, and partially disposed within the free zone). Therefore, torque transfer to anchoring portion **34** is still possible. In FIG. 2F, anchor **12** has transitioned into its non-torque-transfer state, as anchor head **30** has been pulled completely out of socket **24** (i.e., into free zone **25**), thereby rotationally disconnecting the socket from the anchor head. FIG. 2G illustrates that further rotation of driver head **45**, rotates interface **41** and socket **24** (e.g., the entirety of casing **20**), but does not result in further rotation of anchor head **30**, or further screwing of anchoring portion **34** into the tissue. At this point in advancement of anchor **12** into tissue **90**, torque applied by driver **40** to driver interface **41** rotates socket **24** relative to anchor head **30** and anchoring portion **34**. It is therefore hypothesized that the transitioning of anchor **12** from the first state to the second state, in response to resistance from tissue **90** to tissue-facing surface **28**, advantageously reduces a risk of overtightening or damaging the tissue contacted by tissue-facing surface **28**.

(20) At this point, the screwing of anchor **12** into tissue **90** is typically complete, and driver **40** can be removed (FIG. 2H).

(21) As described briefly hereinabove, for some applications, and as shown, anchor **12** comprises compression spring **26** disposed within casing **20** (e.g., within free zone **25**). For some such applications, spring **26** facilitates sustained screwing of anchor **12** into tissue **90** while the anchor transitions from the first state to the second state. Spring **26** can be disposed between anchor head **30** and tissue-facing surface **28**. As anchor head **30** becomes progressively pulled out of from socket **24**, and before the anchor head exits the socket entirely, the anchor head contacts spring **26** (FIG. 2E), such that further rotation of anchor head **30** begins to compress the spring, such that the spring presses tissue-facing surface **28** against tissue **90**. It is hypothesized that, for some applications, spring **26** thereby advantageously increases reliability of anchor **12**, by increasing a likelihood that tissue-facing surface **28** becomes pressed securely against tissue **90** before anchor **12** transitions into its non-torque-transfer state. To facilitate the described function of spring **26**, while the spring is in a relaxed state (e.g., before anchor **12** has been introduced into the subject) an axial height **50** of anchor head **30** can be greater than an axial distance **52** between socket **24** (e.g., a distal end thereof) and spring **26**. In the non-torque-transfer state of anchor **12**, (e.g., once anchor **12** has been screwed into tissue **90**), a combined axial height **54** of anchor head **30** and spring **26** can be similar to, but imperceptibly smaller than, an axial distance **56** between socket **24** (e.g., a distal end thereof) and a distal end of free zone **25**. (Axial distance **56** can, for some applications, be considered the axial height of free zone **25**.)

(22) FIGS. 3A-H show the same sequence as FIGS. 2A-H, mutatis mutandis, but for the anchoring

of anchor **12'**. Anchor **12'** is typically as described for anchor **12**, except that anchor **12'** lacks spring **26**, and can be dimensioned differently in order to accommodate this lack of the spring. Anchor head **30** is typically dimensioned such that, upon the anchor head pressing tissue-facing surface **28** against tissue **90**, the anchor head exits socket **24** (FIG. 3F), thereby transitioning anchor **12'** into its non-torque-transfer state (FIG. 3G).

(23) For some applications, to confer reliability on anchor **12'**, e.g., to reduce a likelihood that anchor **12'** transitions into its non-torque-transfer state before its tissue-facing surface **28** becomes pressed securely against tissue **90**, an axial height **58** of anchor head **30** can be similar to, but imperceptibly smaller than, axial distance **56**.

(24) Reference is made to FIGS. 4A-D, 5A-E, and 6A-B, which are schematic illustrations showing an example tissue anchor system **110**, in accordance with some applications. Reference is also made to FIGS. 7A-C, 8A-E, and 9A-C, which are schematic illustrations showing an example tissue anchor system **210**, in accordance with some applications.

(25) Systems **10**, **110** and **210** have several features in common with each other. Furthermore, components that are identically named between the systems typically share similar features and serve similar functions as each other. For example, each of tissue anchors **112** and **212** comprises a driver interface **141**, **241** shown being engaged by driver head **45** and rotated using driver **40**. As such, the description below of systems **110** and **210** focuses upon features that distinguish each system from system **10**.

(26) Systems **110** and **210** are described as comprising anchor driver **40** (described hereinabove in reference to FIGS. 2 and 3A-H), but each of these systems can optionally comprise a different anchor driver.

(27) For some applications, and as shown in FIGS. 4A and 7A, each crown **138**, **238** comprises a respective housing **119**, **219** which comprises a proximal casing **124**, **224** and a distal casing **126**, **226**. A plurality of grooves **123**, **223** (e.g., grooves **123a** and **123b**, or grooves **223a** and **223b**, respectively) are shown as being defined by distal casing **126**, **226** yet the grooves can optionally be defined by proximal casing **124**, *mutatis mutandis*.

(28) Each crowns **138**, **238** respectively comprises an anchor head **130** or gear **230** that is fixedly coupled via a neck **132**, **232** to an anchoring portion **134**, **234**, having a distal tissue-piercing point **136**, **236**, such that rotation of the anchor head or gear rotates the anchoring portion about a longitudinal axis **ax10**, **ax20**, as described hereinabove in reference to anchor **12**.

(29) Crowns **138** and **238** of tissue anchors **112** and **212** do not utilize a socket in the manner described for crown **38** of anchor **12**. Instead, each of crowns **138** and **238** comprises elements that function together as a slip clutch **140**, **240** that couples (e.g., selectively rotatably couples) their respective driver interface **141**, **241** to the respective anchor head **130** or gear **230** of crowns **138**, **238**.

(30) Selective rotational coupling of interface **141**, **241** to anchor head **130** or gear **230** by way of slip clutch **140**, **240** facilitates transfer of torque from the driver interface to the anchor head, yet limits the transferred torque such that the torque does not exceed a torque threshold. It is hypothesized that using a slip clutch to limit the transferred torque reduces a risk of overtightening the anchor or damaging the tissue. It is further hypothesized that, for some applications, using a slip clutch in this manner may also reduce a risk of under-tightening the anchor, by enabling a surgeon to confidently tighten the anchor without inadvertently overtightening.

(31) As shown in FIGS. 4A-D, slip clutch **140** defined by crown **138** of tissue anchor **112** comprises one or more cantilever pins **122** disposed along a respective groove axis **ax5** (e.g., a first cantilever pin **122a** disposed along a first groove axis **ax5a**, and a second cantilever pin **122b** disposed along a second groove axis **ax5b**), such that each pin is disposed within a respective groove **123** defined by housing **119**. As shown in FIG. 5A, groove **123** comprises a loose portion **135**, within which a free portion **145** of pin **122** is disposed, and a tight portion **133** within which a fixed portion **143** of the pin is disposed (e.g., such that the fixed portion is fixedly coupled to driver

interface **141**). A loose-portion width **d2** of the loose portion can be greater than a tight-portion width **d1** of the tight portion.

(32) In some applications, and as shown in FIG. 5A, each groove axis **ax5** lies on a groove plane that is generally perpendicular to longitudinal axis **ax1** and longitudinally aligned with anchor head **130**. In this way, each of pins **122** is longitudinally aligned with anchor head **130**. In transverse cross-section (e.g., FIG. 5A), anchor head **130** has a non-circular profile, defining a plurality of lateral surfaces (e.g., sides) **144**. Although anchor head **130** is shown as having a square profile, this is not meant to exclude other shapes (e.g., other polygons, such as a hexagon). In a resting state of anchor **112** (e.g., as shown in FIGS. 4B and 4D), each of pins **122** is in contact with a lateral surface **144** of anchor head **130**.

(33) In some applications, and as shown, anchor head **130** is coupled to a bearing **120** that is housed within housing **119** such that the bearing is rotationally coupled to the housing, and rotationally couples the anchor head and anchoring portion **134** to the housing. In some applications, bearing **120** is housed snugly within housing **119** so as to provide smooth rotation with little wobble. In this way, rotation of driver interface **141** rotates housing **119**, yet whether rotation of the housing will rotate bearing **120**, anchor head **130** and anchoring portion **134**, is dependent upon contact between cantilever pins **122** (e.g., lateral surface **144** thereof) and the anchor head—i.e., on slip clutch **140**.

(34) In some applications, rotational coupling of driver interface **141** to anchor head **130** is accomplished via contact between cantilever pins **122** and the anchor head, e.g., by the cantilever pins pressing against lateral surfaces **144** of the anchor head. For example, and as described in more detail hereinbelow, the system can be configured such that application, to interface **141**, of torque below the torque threshold, rotates the housing **119**, pins **122** and anchor head **130** in unison, while the pins remain in contact with lateral surfaces **144** of the anchor head. However, application of torque above the torque threshold will typically cause the anchor head to push against the pins, such that the pins deflect laterally away from longitudinal axis **ax10**, while the pins (and housing **119**) revolve around the anchor head. In this way, torque exceeding the torque threshold may not be transferred to anchor head **130**.

(35) For some applications, and as shown in FIG. 5A, torque is transferred to lateral surface **144** of anchor head **130**, from a portion of pins **122**, e.g., from a torque-applying portion **142** (e.g., a first torque-applying portion **142a**), between fixed portion **143** and free portion **145**—that is in contact with the lateral surface of the anchor head. For some such applications, while screwing anchoring portion **134** into tissue **90**, torque-applying portion **142** comprises a leading end of pin **122**, such that the torque-applying portion revolves ahead of fixed portion **143** while pin **122** revolves about axis **ax10**.

(36) For some applications, and as shown, torque-applying portion **142** is defined merely by virtue of being the portion of pin **122** via which torque is applied to anchor head **130**, rather than being a physical or other distinguishing feature of that portion of the pin.

(37) The rotational arrows in FIG. 5A indicate that torque applied to driver interface **141**, using driver **40**, causes housing **119**, anchor head **130** and anchoring portion **134** to rotate with pins **122**, thereby facilitating screwing the anchor **112** into tissue **90**.

(38) FIG. 5B shows anchor **112** having been screwed into tissue **90**, due to continued application of torque, to anchor interface **141**, in the first direction (e.g., forward torque applied in a forward direction). Screwing anchor **112** into tissue **90** has moved the anchor distally, such that a tissue-facing surface **128** contacts the tissue. At this point, resistance provided by the tissue to further distal movement of tissue-facing surface **128** increases the magnitude of torque required to continue to rotate driver interface **141** above the torque threshold (e.g., increasing the required torque from a first magnitude that is below the torque threshold, to a second magnitude that is above the torque threshold).

(39) As shown in FIG. 5C, anchor head **130** (e.g., lateral surface **144** thereof) begins to deflect pin

122 away from longitudinal axis **ax10**, such that the pins are not entirely parallel to groove axis **ax5**, and slip clutch **140** begins to slip.

(40) In some applications, pins **122** are sufficiently flexible to deflect, while torque is applied to interface **141** at above the torque threshold, and torque-applying portion **142a** contacts anchor head **130** at one end of a cantilever span (e.g., a forward cantilever span **d3**), while fixed portion **143** of the pin is fitted within tight portion **133** at another end of the cantilever span. Thus, the forward cantilever span is typically measured along the pin from (i) torque-applying portion **142a**, to (ii) fixed portion **143**. Flexibility of pin **122** and/or a length of forward cantilever span **d3** can be configured in order to set the torque threshold of slip clutch **140**.

(41) FIG. 5D shows slip clutch **140** having continued to slip, such that pins **122** have deflected further away from longitudinal axis **ax1**, and an inter-pin distance **d5** between points of contact of respective pins **122** with anchor head has increased, as free portions **145** of the pins pivot within loose portions **135** of grooves **123**. At this stage, pins **122** have begun to slip (i.e., revolve) around anchor head **130**, such that continued application of torque at the second magnitude causes driver interface **141** to rotate with respect to the anchor head and anchoring portion **134**—i.e., to rotate without further screwing of the anchoring portion into the tissue.

(42) As shown in FIG. 5E, further rotation of interface **141** allows pins **122** to deflect medially toward their original conformation, and inter-pin distance **d5** is reduced, as the interface and the pins complete a quarter turn since their orientation shown in FIG. 5B.

(43) In certain situations, it may be desirable to remove tissue anchor **112** from tissue **90** (e.g., after having partially or fully screwed the tissue anchor into the tissue). For instance, the surgeon may choose to move the anchor (e.g., to an alternate portion of an implant, and/or to an alternate location of tissue), or to remove the anchor entirely (e.g., due to the anchor no longer being necessary).

(44) As shown in FIGS. 6A-B, removal of anchor **112** from tissue **90** is accomplished by applying reverse torque (i.e., torque in a second rotational direction that is opposite to the first rotational direction) to interface **141**, such that pins **122** revolve about longitudinal axis **ax10** in the second rotational direction, while the pins contact lateral surface **144** of the anchor head.

(45) For some applications, it may be important to ensure that sufficient reverse torque can be applied to unscrew the anchor, despite the original anchoring torque having been limited. Furthermore, in some cases, the surgeon may encounter greater resistance to unscrewing anchor **121**, than that encountered when initially screwing the anchor into tissue **90**. For instance, development of scar tissue at an implantation site of anchor **121** may impede removal of the anchor. In order to facilitate unscrewing of anchor **112** from tissue **90**, some applications of tissue anchor **112** allow more reverse torque than forward torque to be transferred from driver interface **141** to anchor head **130**.

(46) Therefore, for some such applications, reverse torque exceeding the torque threshold (e.g., at the second magnitude) can be transferred from driver interface **141** to anchor head **130**. That is, pins **122** are sufficiently rigid to resist deflection while reverse torque is applied at the second magnitude to interface **141**, and torque-applying portion **142b** contacts anchor head **130** at one end of reverse cantilever span **d4**, while fixed portion **143** of the pin is fitted within tight portion **133** at another end of the reverse cantilever span. For some such applications, while unscrewing anchoring portion **134** from tissue **90**, torque-applying portion **142** comprises a leading end of pin **122**, such that the torque-applying portion revolves ahead of fixed portion **143** while the pin revolves about axis **ax10**.

(47) As shown, the reverse cantilever span is typically measured along the pin from (i) torque-applying portion **142b**, to (ii) fixed portion **143**. Reverse cantilever span **d4** is typically shorter than forward cantilever span **d3**, such that torque-applying portion **142b** is closer to fixed portion **143** when unscrewing anchor **112**, than torque-applying portion **142a** is to fixed portion **143** when screwing the anchor into tissue **90**. It is hypothesized that a magnitude of torque that can be

applied, via pin **122**, from anchor interface **141** to anchor head **130**, is inversely related to the length of the cantilever span, such that a greater magnitude of torque can be transferred along a shorter cantilever span.

(48) However, it may be desirable to limit also the magnitude of reverse torque that can be applied. Therefore, for some applications, slip clutch **140** limits the magnitude of reverse torque that can be applied from interface **141** to anchor head **130** while unscrewing tissue anchor **112**. For such applications, a second torque threshold (i.e., a reverse torque threshold), greater than the first torque threshold, is typically established. Thereby, application of torque at a third magnitude, exceeding the second torque threshold, may cause slip clutch **140** to slip, and driver interface **141** to rotate with respect to anchor head **130** and anchoring portion **134**.

(49) Reference is made again to FIGS. **7A-C**, **8A-E**, and **9A-C**, which show tissue anchor **212** of system **210**. As described hereinabove, tissue anchor **212** shares features with tissue anchor **112**. As such, the description hereinbelow focuses upon features that distinguish anchor **212** from anchor **112**, particularly features of slip clutch **240** which differ from those of slip clutch **140**. For example, slip clutch **240** comprises a gear **230** in place of anchor head **130**, and a cantilever pin **222** in place of pin **122**. Slip clutch **240** is therefore a ratcheting slip clutch in which pin **222** defines a pawl that interacts with gear **230** as described in greater detail hereinbelow.

(50) Similarly to slip clutch **140**, slip clutch **240** facilitates transfer of torque from the driver interface to the anchor head, yet limits the magnitude of torque that can be applied when screwing tissue anchor **212** into tissue **90**, by selectively rotationally coupling driver interface **241** to gear **230**.

(51) Grooves **223** are dimensioned to snugly fit pins **222**, similarly to way that tight portions **133** fit pins **122**, and pins **222** are typically dimensioned such that while fixed portions **243** thereof are disposed within grooves **223**, the pins (e.g., torque-applying portions **242** thereof) are in contact with gear **230** (e.g., a non-circular lateral surface **244** thereof). In a resting state of anchor **212** (e.g., as shown in FIGS. **7B** and **7C**), each of pins **222** is in contact with lateral surface **244** of gear **230**.

(52) For some applications, and similarly to as described hereinabove in reference to anchor head **130** of system **110**, gear **230** is coupled to a bearing **220** that is housed within housing **219** such that the bearing is rotationally coupled to the housing, and rotationally couples the gear and anchoring portion **234** to the housing. For some applications, bearing **220** is housed snugly within housing **219** so as to provide smooth rotation with little wobble. In this way, rotation of driver interface **241** rotates housing **219**, yet whether rotation of the housing will rotate bearing **220**, gear **230** and anchoring portion **234**, is dependent upon contact between pins **222** (e.g., lateral surface **244** thereof) and the gear.

(53) FIG. **8A** shows anchor **212** being screwed into tissue **90**. The rotational arrows in FIG. **8A** indicate that torque applied to driver interface **241**, using driver **40**, causes housing **219**, gear **230** and anchoring portion **234** to rotate, thereby screwing anchoring portion **234** into tissue **90**. Similarly to as described hereinabove in reference to slip clutch **140**, while torque is applied to interface **241** at under the torque threshold, pins **222** revolve about longitudinal axis **ax20** while torque-applying portions **242a** of the pins are in contact with lateral surface **244** of gear **230**, causing the gear and anchoring portion **234** to rotate. However, in contrast that described hereinabove with reference to slip clutch **140**, while screwing anchoring portion **234** into tissue **90**, fixed portion **243** comprises a leading end of pin **222**, such that the fixed portion revolves ahead of torque-applying portion **242** while the pin revolves about axis **ax20**.

(54) FIG. **8B** shows anchor **212** having been screwed into tissue **90**, due to continued application of forward torque to anchor interface **241**. Screwing anchor **212** into tissue **90** has moved the anchor distally, such that a tissue-facing surface **228** contacts the tissue. As described hereinabove in reference to tissue anchor **121**, resistance provided by the tissue to further distal movement of tissue-facing surface **228** increases the magnitude of torque required to rotate driver interface **241** (e.g., from the first magnitude to the second magnitude) to beyond the torque threshold.

(55) As shown in FIG. 8C, gear 230 (e.g., lateral surface 244 thereof) begins to deflect torque-applying portions 242a of pins 222 laterally away from longitudinal axis ax20, such that the pins are not entirely parallel to axis ax5. Pins 222 begin to slip around gear 230, thereby reducing the torque transferred from driver interface 241 to gear 230. FIG. 8D shows gear 230 having further deflected pins 222 (e.g., torque-applying portions 242a thereof) away from longitudinal axis ax20, such that slip clutch 240 has continued to slip around gear 230. At this stage, application of forward torque at the second magnitude causes driver interface 241 and pins 222 to rotate with respect to gear 230 and anchoring portion 234.

(56) As shown in FIG. 8E, further rotation of interface 241 allows pins 222 to deflect medially toward their original conformation, as the interface and the pins complete a quarter turn since their orientation shown in FIG. 8B.

(57) Similarly to as described hereinabove regarding tissue anchor 112 with reference to FIGS. 6A-B, it may be desirable in certain situations to remove tissue anchor 212 from tissue 90 (e.g., after having partially or fully screwed the tissue anchor into the tissue).

(58) As shown in FIGS. 9A-B, removal of anchor 212 from tissue 90 is accomplished by applying reverse torque to interface 241, such that pins 222 revolve about longitudinal axis ax20 in the second rotational direction, while the pins contact gear 230.

(59) However, the manner in which pins 222 of anchor 212 contact gear 230 while revolving about longitudinal axis ax20 in the second rotational direction is different from the manner in which the pins contact the gear while revolving in the first direction. As shown in FIG. 9B, reverse rotation of driver interface 241 can cause some reverse rotation of pins 222 in relation to gear 230 (e.g., “backlash”). As shown, a degree of backlash permitted by reverse rotation of pins 222 is typically limited to less than a quarter turn.

(60) FIG. 9B shows pins 222 having rotated in the second direction until the pins engage gear 230 (e.g., until end-portions 248 are latched into notches 246). In some applications, once end-portions 248 fit into notches 246, backlash is stopped, and pins (e.g., torque-applying portions 242b) transfer torque from interface 241 to gear 230, this time in the second direction.

(61) Thereby, slip clutch 240 may allow reverse torque that exceeds the torque threshold to be transferred from driver interface 241 to gear 230. As shown in FIGS. 9B-C, latching of end-portions 248 of pins 222 into notches 246 facilitates unscrewing anchor 212 from tissue 90, by gear 230 and anchoring portion 234 rotating with driver interface 241, in response to pin 222 revolving about longitudinal axis ax20 in the second direction.

(62) Referring again to FIGS. 1A-9C, the tissue anchors described herein can be used to fasten one tissue to another, and/or to secure another element (e.g., an implant) to tissue. For example, a system for treating a patient can include an implant that is secured to the tissue with any of the tissue anchors described herein.

(63) The tissue anchors described herein can be used for anchoring to cardiac tissue, such as to an atrial wall, a ventricular wall, or a valve annulus of a heart. For some applications, one or more of the tissue anchors described herein can be used to secure, to cardiac tissue, an implant. For some applications, one or more of the tissue anchors described herein can be used to secure, to cardiac tissue, a tether or contraction member (e.g., contraction wire, contraction ribbon, contraction suture, etc.) that is to be tensioned in order to change a shape and/or function of the heart. For example, one or more of the tissue anchors described herein can be used to secure, to ventricular tissue, a tether or contraction member that is also secured to a valve leaflet. For some applications, a plurality of tissue anchors described herein can be used to secure, to a valve annulus, an annuloplasty structure, e.g., with the anchors serving, mutatis mutandis, in place of the anchors described in one or more of the embodiments disclosed in U.S. Pat. No. 9,949,828 to Sheps et al., US Patent Application Publication 2020/0015971 to Brauon et al., PCT Application PCT/IB2020/060044 to Kasher et al. (which published as WO 2021/084407), and/or U.S. Provisional Patent Application 63/147,699 to Shafigh et al., each of which is incorporated herein by

reference. Furthermore, one or more features of the tissue anchors described in these incorporated references can be provided on any of the tissue anchors described hereinabove. For example, the scope of the present disclosure includes modifying any of the anchors described hereinabove to include (e.g., on the crown of the tissue anchor) an eyelet or rotating eyelet, such as described in one of these incorporated references, e.g., in order to slidably couple the anchor to a tether or contraction member that is to be anchored around an annulus of a heart valve and subsequently tensioned in order to perform transluminal annuloplasty. For some applications, an implant comprises a tether/contraction member and one or more of the anchors described herein.

(64) The present invention is not limited to what has been particularly shown and described herein. Rather, the scope of the invention includes both combinations and subcombinations of the various features described herein, as well as variations and modifications thereof that are not in the prior art, which would occur to persons skilled in the art upon reading the foregoing description. The treatment techniques, methods, operations, steps, etc. described or suggested herein can be performed on a living animal or on a non-living simulation, such as on a cadaver, cadaver heart, simulator (e.g., with the body parts, tissue, etc. being simulated), etc.

Claims

1. A system for use with tissue of a subject, the system comprising: a driver; and a tissue anchor, the tissue anchor comprising: an anchoring portion configured to be screwed distally into the tissue by being rotated about a longitudinal axis of the anchor; and a crown, coupled to a proximal portion of the anchoring portion, defining a tissue-facing surface, and comprising: an anchor head fixedly coupled to the anchoring portion, such that screwing the anchoring portion into the tissue moves the anchor head distally along the longitudinal axis toward the tissue, a driver interface, configured to be engaged by the driver, and a socket, fixedly coupled to the driver interface, and shaped to receive the anchor head within the socket, the tissue-facing surface facing distally away from the socket, wherein the tissue anchor has: a first state in which the anchor head is seated snugly within the socket, such that torque applied by the driver to the driver interface rotates the socket, the anchor head, and the anchoring portion, thereby facilitating screwing of the anchoring portion into the tissue, and a second state in which the anchor head is disposed distally from the socket, such that torque applied by the driver to the driver interface rotates the socket relative to the anchor head and the anchoring portion, and wherein the tissue anchor is configured to transition from the first state to the second state, responsively to the anchoring portion having been screwed into the tissue sufficiently deep such that the tissue resists further distal movement of the tissue-facing surface while the screwing of the anchoring portion into the tissue pulls the anchor head distally out of the socket.
2. The system according to claim 1, wherein the driver interface defines a floor that separates the driver from the anchor head while the driver interface is engaged by the driver.
3. The system according to claim 1, wherein the anchor head is shaped such that a transverse cross-section of the anchor head defines a non-circular profile.
4. The system according to claim 3, wherein the anchor head is shaped such that the transverse cross-section of the anchor head defines a plurality of lateral surfaces.
5. The system according to claim 3, wherein the anchor head is shaped such that the transverse cross-section of the anchor head defines at least one of a polygon, a square, and a hexagon.
6. The system according to claim 1, wherein the tissue is tissue of a heart of the subject, and wherein the tissue anchor is transluminally advanceable to the heart.
7. The system according to claim 6, wherein the driver comprises a flexible shaft and a driver head at a distal end of the shaft, such that the anchor driver is transluminally advanceable to the heart.
8. The system according to claim 1, wherein the crown comprises a casing, the casing dimensioned to define: the driver interface, the socket, the tissue-facing surface, and a free zone disposed

between the socket and the tissue-facing surface, wherein, while the anchor is in the second state, the anchor head is disposed within the free zone.

9. The system according to claim 8, wherein the anchor head is rotatable with respect to the socket while the anchor head is disposed in the free zone.

10. The system according to claim 8, wherein the driver comprises a driver head, the driver head shaped to define a shoulder, the shoulder: positioned on a side of the driver head, and dimensioned such that, while the driver interface is engaged by the driver head, the shoulder contacts a proximal surface of the casing.

11. The system according to claim 8, further comprising a spring disposed within the casing, between the anchor head and the tissue-facing surface, wherein the anchor is configured such that while the anchor transitions from the first state to the second state: screwing the anchoring portion into the tissue pulls the anchor head distally out of the socket, compressing the spring.

12. The system according to claim 11, wherein the anchor is configured such that while the anchor transitions from the first state to the second state, screwing the anchoring portion into the tissue pulls the anchor head distally out of the socket, compressing the spring and pressing the tissue-facing surface against the tissue.

13. The system according to claim 11, wherein the anchor is configured such that while the anchor transitions from the first state to the second state, screwing the anchoring portion into the tissue pulls the anchor head distally out of the socket, compressing the spring while the anchor head is: partially disposed within the socket, and partially disposed within the free zone.

14. The system according to claim 1, further comprising an implant, wherein the tissue anchor is configured to secure the implant to the tissue.

15. The system according to claim 14, wherein the implant comprises a tether or contraction member, wherein the tissue anchor is configured to secure the tether or contraction member to the tissue such that applying tension to the tether or contraction member changes a shape and/or size of the tissue.
